

Problem Statement

AI-Powered FAQ System for NADRA

Internship Project | 2025

1. Background

The National Database and Registration Authority (NADRA) is Pakistan's primary identity management body, operating under the Government of Pakistan. NADRA is responsible for issuing and maintaining identity documents for tens of millions of citizens across the country and abroad. Its core services include:

- **CNIC** — Computerised National Identity Card, the primary identity document for Pakistani citizens
- **NICOP** — National Identity Card for Overseas Pakistanis
- **POC** — Pakistan Origin Card, issued to foreign nationals of Pakistani origin
- **CRC / Juvenile Card** — Child Registration Certificate for minors under 18
- **FRC** — Family Registration Certificate, documenting family composition

Together, these services affect virtually every Pakistani citizen at some point in their life — from birth registration to overseas residency. Given the scale of NADRA's operations and the diversity of the population it serves, citizen queries related to these services are extremely high in volume and highly repetitive in nature.

2. The Problem

Despite the critical nature of NADRA's services, citizens frequently struggle to find clear, reliable answers to basic questions about them. Common queries — such as which documents are required for a CNIC renewal, what the current fee structure is, or how long processing takes — require users to navigate through multiple web pages, embassy portals, and unofficial sources, often arriving at outdated or conflicting information.

NADRA's existing support channels, including its helpline, complaint portal, and social media presence, are burdened with a high volume of these routine, answerable queries. This places unnecessary load on human support staff, increases wait times for citizens with genuine issues, and results in a poor overall service experience — particularly for overseas Pakistanis and citizens with limited digital literacy.

The core challenge is that there is currently no intelligent, conversational system capable of answering citizen queries about NADRA services accurately, instantly, and in plain language — without requiring citizens to dig through official documentation themselves.

3. Proposed Solution

This project proposes the development of an AI-powered question-answering system specifically designed to handle citizen queries about NADRA's services. The system will be built using a technique known as Retrieval-Augmented Generation (RAG), which enables an AI model to answer questions by referring to a curated, trusted knowledge base drawn directly from NADRA's official information — rather than relying on general internet knowledge.

This approach ensures that the system's responses are always grounded in accurate, NADRA-specific information. If a question falls outside the scope of the system's knowledge, it will clearly indicate this rather than risk providing an incorrect answer — a critical requirement for a government services context.

The system will be capable of answering questions about required documents, eligibility criteria, fee structures, processing timelines, and renewal procedures across all five of NADRA's core services.

4. Objectives

- Collect and organise accurate, up-to-date information about all five NADRA services into a structured knowledge base
- Develop an AI model that can understand citizen questions and retrieve the most relevant information from that knowledge base
- Ensure the system responds strictly based on verified NADRA information, with no fabricated or misleading answers
- Evaluate the accuracy and reliability of the system's responses before any public deployment
- Lay the foundation for a future citizen-facing interface through which the system will be publicly accessible

5. Scope

In Scope	Out of Scope (Current Phase)
• FAQ answering for CNIC, NICOP, POC, CRC, and LRA • Required documents, eligibility, fees, and timelines • English language support • Accuracy testing and evaluation	• Application status tracking • Complaint registration or grievance handling • User-facing frontend / interface • Urdu language support (future stretch goal) • Integration with NADRA internal systems

6. Phased Delivery

Phase 1 — Current

Build and evaluate the core AI model. This includes assembling the NADRA knowledge base, developing the question-answering pipeline, and rigorously testing the system for accuracy and

reliability.

Phase 2 — Future

Develop and integrate a user-facing interface, making the system accessible to citizens through a web or mobile application.

7. Expected Outcomes

- A verified, well-organised knowledge base covering all NADRA services
- An AI question-answering model that responds accurately to citizen queries
- A tested and documented system ready for integration into a future user-facing product
- A foundation for expanding the system to handle Urdu queries in future iterations

This document is subject to revision as project requirements are further defined.